TRY THESE

- Find the value of 6 + 3 + 8 and 3 + 6 + 8
  i) Are they same?
  ii) Is there any other way of arranging these three numbers?
- Find the value of 5 × 2 × 6 and 2 × 5 × 6
  i) Are they same?
  ii) Is there any other way of arranging these three numbers?
- Is 7 – 5, the same as 5 – 7? Why?
- What is the value of (15 – 8) – 6? Is it the same as 15 – (8 – 6)? Why?
- What is 15 ÷ 5? Is it the same as 5 ÷ 15? Why?
- What is the value of (100 ÷ 10) ÷ 5? Is it the same as 100 ÷ (10 ÷ 5)? Why?

1.10 Properties of Whole Numbers

The properties of numbers are important facts to be remembered,which helps to do arithmetic calculations more precisely and to avoid mistakes.

1.10.1 Commutativity of addition and multiplication

When two numbers are added (or multiplied), the order of the numbers does not affect the sum (or the product). This is called commutativity of addition (or multiplication).

Observe the given facts:

(i) 43 + 57 = 57 + 43

(ii) 12 × 15 = 15 × 12

(iii) 35,784 + 48,12,69,841 = 48,12,69,841 + 35,784

(iv) 39,458 × 84,321 = 84,321 × 39,458

Such facts are called as equations. In each of the above equations, the answers on both the sides are same. Finding the answer for the third and fourth equations takes more time. But, these equations are meant to convey the properties of numbers. The third equation is correct by commutativity of addition and the fourth equation is correct by commutativity of multiplication.

There is a nice pictorial way of understanding commutativity of multiplication. If we have 5 rows of stars, each with 4 stars, we can draw the total of 20 stars as a rectangle (5 × 4 = 20). See Fig.1.2 below. Now rotate the rectangle given in Fig.1.2 (a) to get the Fig.1.2(c) as given below. It is the same rectangle. It has exactly the same total number of stars, 20. But now we have 4 rows of stars, each with 5 stars! That is, 5 × 4 = 4 × 5.

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